

Machine Learning – Think & Answer

Solutions to 30 Conceptual Questions

Focus: Linear Regression, Cost Optimization, Gradient Descent, Overfitting, Lasso, Ridge, Logistic Regression and KNN

A. Linear Regression & Error

1. House price prediction error

Answer: Predicted price = ₹50 lakhs and actual price = ₹55 lakhs. Prediction error = Actual – Predicted = ₹5 lakhs. During training, the model should adjust its parameters to reduce prediction errors and lower the overall cost.

2. Large errors for many training examples

Answer: I would try to minimize the cost function. I would check the features and data quality, then adjust model parameters using optimization such as Gradient Descent. Useful features may need to be added and irrelevant or problematic data investigated. The expected result is lower overall error.

3. Perfect training performance but poor new-data performance

Answer: This is most likely overfitting. The model has learned the training data too closely, including noise or unusual patterns. Regularization, feature selection, a simpler model, more data, or cross-validation can improve generalization.

4. Why not keep adding features?

Answer: Adding features can reduce training error, but irrelevant, redundant, or noisy features can increase complexity and cause overfitting. Instead, select useful features and control model complexity.

B. Cost Function & Optimization

5. Cost 100 versus cost 25

Answer: Model B is currently better because its cost is lower. Training should try to minimize the cost function, while also checking validation/test performance.

6. Gradient Descent and the mountain

Answer: Treat the cost function as a landscape. Gradient Descent determines the direction of increasing cost and moves in the opposite direction, repeatedly updating parameters until a low-cost point is reached.

7. Extremely large learning rate

Answer: The updates can become too large. The algorithm may jump over the minimum, oscillate, fail to converge, or even make the cost increase.

8. Extremely small learning rate

Answer: The model takes very small steps. Training may be stable but extremely slow, requiring many iterations to reach a good solution.

9. Cost 100 → 60 → 30 → 15 → 14

Answer: The model is learning because the cost is steadily decreasing. The improvement is becoming smaller, suggesting that training is approaching a minimum or a point of diminishing returns.

C. Overfitting & Regularization

10. Training accuracy 99%, testing accuracy 65%

Answer: I would not trust the model immediately. The large gap strongly suggests overfitting. I would consider regularization, feature selection, a simpler model, more data, and cross-validation.

11. Almost zero training error means best model?

Answer: No. A model can memorize the training data and still perform poorly on unseen data. The important question is how well it generalizes to validation/test data.

12. Overfitting: complex or simpler?

Answer: I would generally make the model simpler or constrain it. Feature selection, regularization, more data, or reducing model complexity can help improve generalization.

13. 50 features with many unimportant features

Answer: This can create unnecessary complexity, increase overfitting risk, increase computation, reduce interpretability, and introduce noise. Feature importance or regularization can help.

D. Lasso & Ridge Regression

14. Why is Lasso useful with many features?

Answer: Lasso can shrink some coefficients exactly to zero. Those features effectively stop contributing to the prediction, so Lasso can perform feature selection and produce a simpler, more interpretable model.

15. 100 features but only 10 important

Answer: Lasso would be a strong choice when feature selection is desired because it can set unnecessary coefficients to zero. Ridge normally shrinks coefficients without making them exactly zero.

16. Why does shrinking coefficients help?

Answer: Large coefficients can make a model sensitive to changes in input data. Ridge discourages excessively large coefficients, which can reduce model complexity, overfitting, and instability.

17. Overfitting because coefficients are extremely large

Answer: Ridge can help by adding a penalty for large coefficients. It encourages smaller coefficients, reducing model complexity and potentially improving test performance.

18. Lasso removes an important feature

Answer: The model may lose useful predictive information, causing accuracy to decrease or underfitting to occur. The regularization strength and validation performance should therefore be checked.

E. Logistic Regression

19. Probability 0.82 with thresholds 0.5 and 0.9

Answer: With threshold 0.5, $0.82 \geq 0.5$, so predict Class 1 (for example, Buy). With threshold 0.9, $0.82 < 0.9$, so predict Class 0 (for example, Don't Buy).

20. Student pass/fail

Answer: Logistic Regression is more appropriate because Pass/Fail is a categorical classification problem. It produces a probability between 0 and 1 that can be converted into a class using a threshold.

F. K-Nearest Neighbors (KNN)

21. K = 1 and an outlier

Answer: With $K=1$, the single nearest training point determines the prediction. If that point is an outlier, it can strongly influence or completely change the prediction. $K=1$ is therefore highly sensitive to noise.

22. Increase K from 3 to 50

Answer: The model becomes less sensitive to individual points because many neighbors contribute to the decision. However, an excessively large K can make the model too generalized and may cause underfitting.

23. Why feature scaling matters

Answer: KNN uses distance. If Age ranges from 18–60 but Income ranges from \$20,000–\$2,00,000, income can dominate the distance calculation. Scaling methods such as standardization or Min-Max normalization make the features contribute more appropriately.

24. KNN training 95%, testing 70%

Answer: This suggests possible overfitting. I would investigate whether K is too small and test several K values using validation or cross-validation to find a better balance between sensitivity and generalization.

G. Final Challenge – Model Selection

25. Linear Regression low training error; Lasso/Ridge moderate training error

Answer: I would not automatically choose Linear Regression. The main goal is unseen-data performance. I would compare validation/test error, cross-validation results, and model complexity. The needed additional information is especially validation/test performance.

26. KNN 92%, Logistic Regression 89%

Answer: No. Accuracy alone is not enough. I would check test performance, cross-validation, precision, recall, F1-score, confusion matrix, class balance, feature scaling, dataset size, and computational requirements.

27. Training error decreases while testing error increases

Answer: This is a classic sign of overfitting. The model is becoming increasingly specialized to training data. I would stop around the point where validation performance is best or use regularization, early stopping, a simpler model, or feature selection.

28. Regularization raises training error but improves testing performance

Answer: No. This can be a good result. A small increase in training error accompanied by a decrease in testing error means regularization has reduced overfitting and improved generalization.

29. Highly correlated features — Ridge or Lasso?

Answer: I would investigate both. Ridge is often a strong candidate for correlated predictors because it shrinks coefficients rather than aggressively eliminating features. I would examine correlations, feature-selection needs, and cross-validation performance before deciding.

30. Choosing among Linear Regression, Logistic Regression, KNN, Lasso, and Ridge

Answer: I would ask: What type of problem is it—regression, classification, or clustering? What is the target variable? How much data is available? What types of features are present? Do features require scaling? Is overfitting a concern? Is feature selection needed? Are features highly correlated? Is interpretability important? What evaluation metrics are appropriate? The algorithm should match the problem and should be selected based on validation/unseen-data performance rather than training error alone.